



Class: XI 'Science'

Time: 3hrs.

First-Term Examination 2080

SET - A

Subject: Physics '1011'

F.M.: 7%

P.M.: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions:

GROUP 'A' [11×1=11]

Multiple choice questions

- 1) If energy E , velocity V and force F are taken as fundamental physical quantities, the dimensional formula of mass is
 a) $[F^0 E V^{-2}]$ b) $[F E V^2]$ c) $[F E V]$ d) $[F E^0 V^{-2}]$
- 2) If sum of two-unit vectors is a unit vector, then the magnitude of their difference is
 a) $\sqrt{2}$ b) 2 c) $\sqrt{3}$ d) $\sqrt{5}$
- 3) Which of the following cannot be resultant of the vectors of the magnitude 9 and 3?
 a) 3 b) 6 c) 12 d) 16
- 4) A force of 20 N is applied at an angle of 60° to the horizontal. What is the vertical component of this force?
 a) 20N b) 10N c) 17.32N d) 15N
- 5) The maximum and minimum magnitude of the resultant of two given vectors is 7 units and 1 unit respectively. If these two vectors are at right angles to each other, the magnitude of resultant is
 a) 4 b) 5 c) 6 d) 7
- 6) A certain vector in xy plane has an x-component of 12 m and y-component of 8m. It is then rotated in xy plane so that its x-component is halved. Then its new y component is approximately
 a) 14m b) 10.12m c) 2.14 m d) 13.11m
- 7) An ant moves towards the plane mirror with speed of 3m/s and the mirror is moved towards the ant with the same speed. What is the velocity of ant with respect to its image?
 a) 3 m/s b) 18 m/s c) 9 m/s d) 12 m/s
- 8) The focal length of concave mirror is f and the distance from the object to the principal focus is x . The ratio of size of the image to the size of the object is
 a) $\frac{f+x}{f}$ b) $\frac{x}{f}$ c) $\frac{f}{x}$ d) $\frac{f}{f+x}$
- 9) A point object is placed on the axis of the concave mirror at a distance of 60 cm from the focal point of the mirror. Its image is formed at the point of object then focal length of the mirror is
 a) 60 cm b) 120 cm c) 30 cm d) 15 cm
- 10) Two equally charged, identical metal spheres A and B repel each other with force F . The spheres are fixed with a distance r between them. Third identical but uncharged sphere is brought in contact with A and then placed at midpoint of line joining A and B. The magnitude of net electrostatic force on sphere A is
 a) $3F/2$ b) F c) $2F$ d) $F/2$
- 11) On a new linear scale of thermometer W scale, the freezing and boiling point of water are $39^\circ W$ and $239^\circ W$ respectively. What will be the temperature on the new scale corresponding to the $39^\circ C$ on the Celsius scale
 a) $200^\circ C$ b) $78^\circ C$ c) $139^\circ C$ d) $117^\circ C$

GROUP 'B' [5×5=25]

Short answer questions:

- a) What are the limitations of dimensional analysis? [1]
- b) Check the correctness of the physical relation $\rho = \frac{3g}{4\pi r G}$ where symbols carry their usual meanings. [2]
- c) Find the value of 20 J on the system based on 20 cm, 1kg and $\frac{1}{2}$ minute as fundamental units. [2]

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- 13) a) State and use parallelogram law of vector addition to find the magnitude and direction of the resultant of two vectors inclined at an angle θ . [3]
 b) Two forces F_1 and F_2 act at a point. F_1 has magnitude 8N and acts at an angle of 30° with the positive x-axis. F_2 has a magnitude of 6N and acts at an angle of 150° with positive x-axis. Find the magnitude and direction of resultant force. [2]
- 14) a) Derive a relation between focal length and radius of curvature of a spherical mirror. [2]
 b) A small candle 2.5 cm in size is placed 27 cm in front of a concave mirror of radius of curvature 36 cm. At what distance from the mirror should a screen be placed in order to receive a sharp image? Describe the nature & size of the image. [3]
- 15) a) Derive a mirror formula for concave mirror and mention the sign convention. [3]
 b) A concave mirror forms an image 3 times as large as an object placed 0.3 m from it. Calculate its focal length. [2]
- 16) a) Define the coefficient linear, superficial and cubical expansions. Derive relations between them. [4]
 b) Why a thick glass cracks when boiling water is poured in it? [1]
- 17) a) Pendulum clocks generally go fast in winter and slow in summer. Why? [1]
 b) A pendulum clock loses 12 Sec. a day if the temperature is 40°C and gains 4 Sec. a day if the temperature is 20°C . Find the temperature at which the clock will show correct time and coefficient of linear expansion of the metal of pendulum shaft. [4]
- 18) a) State and explain the principle of superposition of Coulomb's force. [2]
 b) Two-point charges of $10\ \mu\text{C}$ and $-40\ \mu\text{C}$ are separated by distance of 13 cm. Find the point along the line joining them at which net electrostatic force on any charge is zero. [3]
- 19) a) Define dielectric constant. Why its value is always greater than one? [2]
 b) Four equal charges q are placed at the four corners of a square of length a . Find the magnitude of the net force acting on one of them due to other charges and draw diagram to show the direction of force on that charge. [3]

GROUP 'C' [3×8=24]

Long answer questions:

- 20) a) Explain why vector addition is ~~not~~ commutative, providing a real-life example to support your reasoning. [2]
 b) What is meant by resolution of vectors? How will you resolve a vector in a plane in two mutually perpendicular directions? [3]
 c) A pilot flies 200 km at 60° north of east, then 150 km at 30° south of west and finally 100km at 45° south of east. Calculate the magnitude and direction of the resultant displacement. [3]
- 1) a) Why does a solid expand on heating? [1]
 b) Explain an experiment to determine the coefficient of linear expansion of a solid. [3]
 c) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder at 15°C is 8.0 cm and there is an all-round clearance of 0.05 mm between the piston and the cylinder of the wall. Find the temperature at which the two will exactly fit in.
 ($\alpha_{\text{steel}} = 1.2 \times 10^{-5}\ ^\circ\text{C}^{-1}$ and $\alpha_{\text{aluminium}} = 1.6 \times 10^{-5}\ ^\circ\text{C}^{-1}$) [4]
- a) State and explain Coulomb's law in electrostatics. [4]
 b) Why coulomb's law is applicable to stationary charges only? [3]
 c) Define 1 coulomb charge on the basis of Coulomb's law. [1]
 d) Two charges 10^{-6}C and 10^{-7}C attract each other with a force of 400N. Calculate the distance between them if they are in
 i) air and [1]
 ii) water (relative permittivity of water is 80) [3]

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